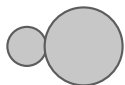
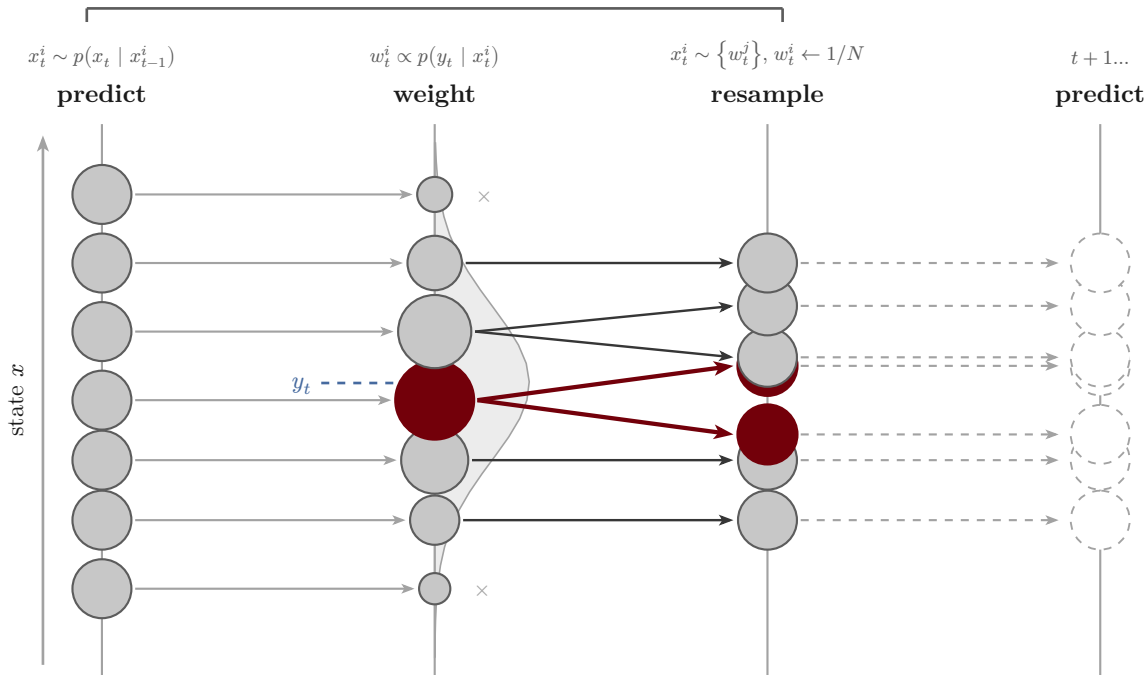


Particle filter · sequential Monte Carlo

one SMC step: $\hat{p}(x_{t-1} \mid y_{1:t-1}) \rightarrow \hat{p}(x_t \mid y_{1:t})$



dot size = weight w_t^i

x = killed

→ resample: duplicate heavy particle

$$\hat{p}(x_t \mid y_{1:t}) = \sum_{i=1}^N w_t^i \delta(x_t - x_t^i) \quad \sum_i w_t^i = 1 \quad N = 7 \text{ particles}$$